3rd-Semester CSE Java Web-Based Project Evaluation

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Online Learning Management System (LMS)

A genuine Servlet-based Java web application built strictly with Jakarta Servlets 6.0, direct JDBC 4.x, Apache Tomcat 10.1, and MySQL. Zero ORM or Spring Boot abstractions.

CRITERION 1

Problem & Design

8 Marks Allocated

CRITERION 2

Core Java Concepts

10 Marks Allocated

CRITERION 3

Database & JDBC

8 Marks Allocated

CRITERION 4

Servlets & Web

7 Marks Allocated

Stack: Java 17+, Jakarta Servlets 6.0, Tomcat 10.1, JDBC, HikariCP, React 18, Tailwind CSS • Status: 39/39 Automated Tests Passed

ACADEMIC CONTEXT

Problem Statement & Educational Motivation

The Educational Problem

- **Fragmented Systems:** Course management, materials, student submissions, and grading are often dispersed across disparate tools, causing high friction for faculty and learners.
- **Security & Privacy Gaps:** Students must never see peers' grades, instructors must not modify other instructors' courses, and public users cannot elevate permissions.
- **Manual Assessment Friction:** Faculty spend valuable time manually grading multiple-choice questions without instant automated feedback.

The Engineering Solution

- **Strictly Grounded Engineering:** Implemented without Spring Boot or ORM (Hibernate) so students genuinely master the fundamental socket, servlet, and JDBC mechanisms.
- **Decoupled 4-Tier Architecture:** Clear separation between React presentation, Jakarta Servlet controllers, Service business rules, and JDBC DAOs.
- **Enterprise Standards:** Complete 3NF relational normalization, 100% PreparedStatements, explicit JDBC transactions, and BCrypt security.

SYSTEM DESIGN

4-Tier Decoupled Layered Architecture

TIER 1: PRESENTATION TIER — Browser / Client

React 18 + TypeScript + Tailwind CSS SPA

Single-page interface with tailored dashboards for Students, Instructors, and Admins. Includes an Evaluator Demo Bar for instant 1-click persona switching.

TIER 2: SERVLET CONTROLLER TIER — Apache Tomcat 10.1 (Port 8080)

Jakarta Servlets 6.0 & BaseServlet Base Class

14 Servlet endpoints handling RESTful HTTP verbs (doGet, doPost, doPut, doDelete). Centralized JSON dispatch and error translation with CorsFilter and SecurityUtils session guards.

TIER 3: BUSINESS SERVICE TIER — Pure Java Interfaces & AppContext

13 Domain Services & Explicit JDBC Transaction Demarcation

Enforces business policies (duplicate enrollment checks, instructor course ownership, grade bounds). Demarcates ACID boundaries using direct JDBC setAutoCommit(false), commit(), and rollback().

TIER 4: DATA ACCESS & STORAGE TIER — HikariCP 5.1 & MySQL 8.x / Embedded H2

11 JDBC DAOs & 14 Relational Tables (3NF Normalized)

100% Parameterized PreparedStatements for SQL injection immunity. Try-with-resources resource cleanup. Automatic failover to embedded H2 for zero-config evaluation.

EVALUATION RUBRIC: 10 MARKS

Core Java Concepts Implementation

1. Encapsulation & Composition

- All model properties are `private` with validated public getters/setters.
- Domain composition mirrors real entities: Course contains Modules; Module contains Lessons; Course contains Assignments and Quizzes.

2. Inheritance & User Hierarchy

- Abstract base class `User` contains shared authentication and profile state.
- Specialized subclasses: `StudentUser` (studentId, major), `InstructorUser` (department), and `AdminUser` (adminLevel).

3. Runtime Polymorphism & Strategy

- Subclasses override abstract methods `getRoleDashboardUrl()` and `hasAdministrativePrivileges()`.
- Strategy pattern via `GradingStrategy` interface (`StandardGradingStrategy` vs `WeightedGradingStrategy`).

4. Generics, Interfaces & Exceptions

- 11 DAO and 13 Service interfaces completely decouple contracts from storage.
- Custom `LmsException` hierarchy mapped centrally to HTTP status codes.
- Strict type-safe collections with generics: `List<Course>`, `Map<String, Object>`, `Optional<T>`.

EVALUATION RUBRIC: 8 MARKS

Direct Database Integration (JDBC 4.x)

PreparedStatement & Resource Safety

100% PreparedStatements: All SQL queries bind dynamic values via positional placeholders (`?`). Dynamic string concatenation is completely barred, ensuring 100% SQL injection immunity.

Try-With-Resources: Every database operation employs AutoCloseable try blocks, ensuring connections, statements, and result sets are released even if exceptions occur.

HikariCP Connection Pool: Maintains a high-throughput connection pool with sub-millisecond acquisition latency.

Generated Key Retrieval: Using `Statement.RETURN_GENERATED_KEYS` to bind child rows immediately without secondary lookups.

Deadlock-Free ACID Transactions

```
// TransactionManager Helper (Eliminates FK Deadlock)
TransactionManager.execute(conn -> {
  // 1. X-Lock FIRST + Capacity Enforcement
  courseDao.incrementEnrolledCount(conn, courseId);
  // 2. Child Insert under Parent X-Lock
  enrollmentDao.enroll(conn, enrollment);
  return enrollment;
});
// 3. Post-commit best-effort notification
notificationService.sendNotification(...);
```

Verified through automated test `ServiceTransactionTest.java` (Capacity check, rollback suppression, and drop).

EVALUATION RUBRIC: 7 MARKS

Jakarta Servlets 6.0 & Web Integration

14 Jakarta Servlets

Full suite of servlets mapped under `/api/*`: Auth, User, Course, Module, Lesson, Enrollment, Assignment, Submission, Quiz, Progress, Notification, Analytics, AuditLog, and SystemSetting.

BaseServlet Architecture

Provides standardized JSON request parsing and uniform response wrapping via Jackson Databind. Centralized error handler maps domain exceptions to HTTP 400, 401, 403, 404, 500.

Session Security & Filters

- Standard Jakarta `HttpSession` state tracking.
- `SecurityUtils` role enforcement.
- Invalidation on logout.
- `CorsFilter` & `EncodingFilter` for UTF-8.

RELATIONAL SCHEMA

Normalized Relational Schema (14 Tables in 3NF)

DOMAIN 1

Identity & Governance

- `users` (Auth, Roles, Profile)
- `notifications` (Alerts, Read Status)
- `audit_logs` (Actions, IPs, Timestamps)
- `system_settings` (Key-Value Configs)

DOMAIN 2

Curriculum & Content

- `courses` (Codes, Titles, Instructor FK)
- `modules` (Syllabus Chapters, Orders)
- `lessons` (Content, Durations, Video)
- `lesson_progress` (Completions)

DOMAIN 3

Assessments & Quizzes

- `enrollments` (Unique: user_id + course_id)
- `assignments` (Due Dates, Max Points)
- `submissions` (Student Text, Grades)
- `quizzes`, `questions`, `attempts`

Evaluator Failover Guarantee: HikariCP is pre-configured with zero-configuration fallback: if a local MySQL instance is not active, the system seamlessly initializes an embedded H2 database in MySQL mode and executes full schema creation and seed scripts automatically.

CURRENT MILESTONES

Current Implementation Status & Live Verification

Jakarta Servlet Backend

100% Operational

14 Servlets, 13 Services, 11 DAOs

React 18 Frontend

100% Operational

Student, Instructor & Admin Portals

Automated JUnit 5 Tests

39 / 39 Passed

0 Failures | 0 Errors | 0 Skipped

Embedded Tomcat Server

HTTP 200 OK

Active on http://localhost:8080

Deployable WAR Package

online-lms.war

Ready in target/ directory

Rubric Marks Alignment

33 / 33 Marks

Fully Satisfied & Demonstrable

QUALITY ASSURANCE

Testing Performed & Verification Results

CoreJavaOopTest (8 Tests)
Validates private encapsulation, Student/Instructor/Admin inheritance, polymorphic strategy scoring, Comparable sorting, and custom LmsException.

PASSED

JdbcDaoPersistenceTest (6 Tests)
Validates direct PreparedStatement execution, user password hash lookups, course insertions, SQL unique constraint handling, and connection lifecycle.

PASSED

ServiceTransactionTest (12 Tests)
Validates TransactionManager execution, rollback suppression (addSuppressed), 20-thread concurrency test (0 oversells), conditional capacity enforcement, and symmetric dropCourse.

PASSED

ServletRequestHandlingTest (13 Tests)
Tests BaseServlet JSON request parsing, response serialization, AuthFilter role-based access control, and proper HTTP error code translation (400, 401, 403, 404, 409, 415).

PASSED

ROADMAP

Future Scope & Enhancements

1. WebRTC Live Classrooms

Integrate browser-based video lectures and collaborative whiteboarding directly within course modules using WebSockets and WebRTC peer channels.

2. Automated Plagiarism Detection

Build an automated n-gram fingerprinting engine to compare student text submissions and programming assignments against peer submissions.

3. AI Quiz & Question Generator

Enable instructors to upload PDF lecture notes and generate 4-option MCQ quizzes automatically using AI summarization pipelines.

4. Native Mobile Application

Deliver an Android/iOS mobile application communicating with the exact same Jakarta Servlet REST API endpoints and session cookies.

EVALUATOR GUIDE

Evaluator Viva Demonstration Walkthrough

Student Journey

1. Click `Demo: Student` in top bar.
2. Browse courses and click **Enroll**.
3. Open curriculum, watch lesson, and complete topics.
4. Take the MCQ quiz and view instant calculated score.

Instructor Journey

1. Click `Demo: Instructor` in top bar.
2. View owned courses and create new course.
3. Review student submissions and input grade & feedback.
4. Add quizzes and multiple choice questions.

Administrator Journey

1. Click `Demo: Admin` in top bar.
2. Inspect system metrics and active user roster.
3. Audit real-time security events in Audit Logs.
4. Toggle platform registration and maintenance policies.

Running on: http://localhost:8080/index.html

Ready for Committee Review

Conclusion & Review 1 Viva Defense

The Online Learning Management System satisfies all requirements of the 3rd-Semester CSE Java Web-Based Project marking rubric (33/33 Marks). The application is fully functional, thoroughly tested, and demonstrable live.

ACADEMIC DELIVERABLES COMPLETED:

- ✓ Full Working Codebase
- ✓ 39/39 Unit & Integration Tests
- ✓ System Architecture Diagram
- ✓ Setup & Execution Guide
- ✓ Review 1 Presentation (PPTX & HTML)
- ✓ 14 Normalized Relational Tables
- ✓ Deployable WAR Archive
- ✓ Database ER Diagram
- ✓ Viva Preparation Q&A Guide
- ✓ Live Server on Port 8080

Thank you. We welcome your questions.